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Excessive Serum Concentrations of Acyclovir and Neurotoxicity

Acyclovir is the preferred drug for treating varicella-zoster and herpes simplex virus infections [1, 2]. Laskin, in an extensive review [3], stated that "acyclovir appears to have a very large therapeutic index." Adverse effects reported are phlebitis, reversible renal and hepatic dysfunctions, and, rarely, neurotoxicity. We report a case of remarkably high serum concentrations of acyclovir and reduced drug clearance with associated neurotoxicity in a patient in whom the serum creatinine level suggested drug clearance would not require dosage adjustment.

Case Report

A 17-y-old girl was hospitalized at St. Jude Children's Research Hospital with a one-day history of painful mouth sores and decreased oral intake. She had a metastatic ovarian germ cell tumor that had been treated for two years with combination chemotherapy and cobalt irradiation. During the week before admission, she had received chemotherapy with cisplatin and VP-16 (etoposide). Examination on admission revealed a temperature of 36.4 C, severe oral mucositis, and mild jaundice. Weight and body surface area (BSA) were 38.6 kg and 1.3 m^2, respectively. Initial laboratory findings were as follows: white blood cell count (WBC), 100/mm^3 with 100% lymphocytes; hemoglobin, 9.4 g/dL; platelets, 27 000/mm^3; total bilirubin, 1.9 mg/dL (direct, 1.1 mg/dL); chloride, 91 mEq/L; potassium, 3.0 mEq/L; and serum alanine aminotransferase, 68 U/L. The levels of blood urea nitrogen (BUN) and creatinine were 35 mg/dL and 2.5 mg/dL, respectively, and were considered elevated secondary to dehydration, hydronephrosis in the left kidney, ureteral obstruction, and nephrotoxic chemotherapy. The remaining blood chemistry tests were normal. Initially, treatment consisted of intravenous fluids, parenteral nutrition, and morphine sulfate for pain. By day 3 the levels of serum creatinine and BUN had declined to 1.7 mg/dL and 19 mg/dL, respectively. The next day she became transiently hypotensive (blood pressure, 68/22 mm Hg), febrile, and slightly disoriented and suffered tachycardia and tachypnea. Cultures of blood were obtained, and therapy with dopamine, amikacin (600 mg/m^2 per d, every 6 h), vancomycin (1.2 g/m^2 per d, every 6 h), and ticarcillin (9 g/m^2 per d, every 6 h) was initiated. *Klebsiella pneumoniae* and *Escherichia coli* were isolated from the cultures of blood and were susceptible to amikacin and ticarcillin. Rates of amikacin and vancomycin clearance were decreased, and the doses were adjusted to maintain appropriate drug concentrations. Thereafter, drug concentrations remained in the therapeutic range.

On hospital day 5 the patient's blood pressure had returned to normal (without dopamine), and she was alert

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and oriented. A culture taken from the patient's mouth, at admission, yielded herpes simplex virus. The oral mucositis had been progressive, and acyclovir therapy, 500 mg/m² (16.8 mg/kg) every 8 h, was begun. The levels of serum creatinine and BUN were 1.5 mg/dL and 28 mg/dL, respectively. At this time she was also receiving oxacillin, cefotaxime, amikacin (modified dose), trimethoprim-sulfamethoxazole, and morphine sulfate. Approximately 2 h after the second dose of acyclovir, she had slurred speech and was disoriented. These neurological abnormalities progressed to include myoclonic jerks, tremors, hallucinations, and muscle fasciculations involving predominantly the tongue and face. Morphine sulfate therapy was discontinued 2 h after the first symptoms were noted. Except for the minimally elevated levels of BUN and creatinine, elevated bilirubin (6.1 mg/dL), and decreased magnesium (0.7 mEq/L), the remaining serum chemistry levels were normal. Correction of the serum Mg^{++} with intravenous $MgSO_4$ did not correct the neurological abnormalities.

Acyclovir therapy was stopped after six doses. Over the next 31 h, serum concentrations of acyclovir were measured on five occasions (figure 1). The patient's concentration of acyclovir at 5.6 h after her last dose was 190 μ*M* and by 31 h had declined to 25 μ*M*. Within 24 h there was marked improvement in the neurological dysfunction. A computed tomographic scan of the brain was normal. Electroencephalogram showed diffuse slowing and presence of triphasic waves consistent with a diffuse encephalopathy from drug toxicity. Over the next two to three days the neurological abnormalities resolved.

A two-compartment model accomodating variable doses and dose intervals was fit to the measured acyclovir concentrations by using weighted, nonlinear, least-squares regression [4]. The pharmacokinetic parameters found were a clearance of 28 mL/min per 1.73 m², a steady-state volume of distribution of 10.1 L/m², and an elimination half-life of 9.3 h.

Two weeks later there was a severe recurrence of the herpetic mucositis. On the basis of pharmacokinetic data from the previous course of acyclovir and to obtain a peak level of 60–80 μ*M* and a trough concentration of <20 μ*M*, we started the patient on an initial dose of 400 mg (269 mg/m²), followed by 250 mg (185 mg/m²) eveyr 24 h. At this time her levels of BUN and creatinine were 15.2 mg/dL and 1.1 mg/dL, respectively. She received 10 d of acyclovir therapy and had good clinical results and no evidence of neurological toxicity. Serum concentrations of acyclovir were not measured.

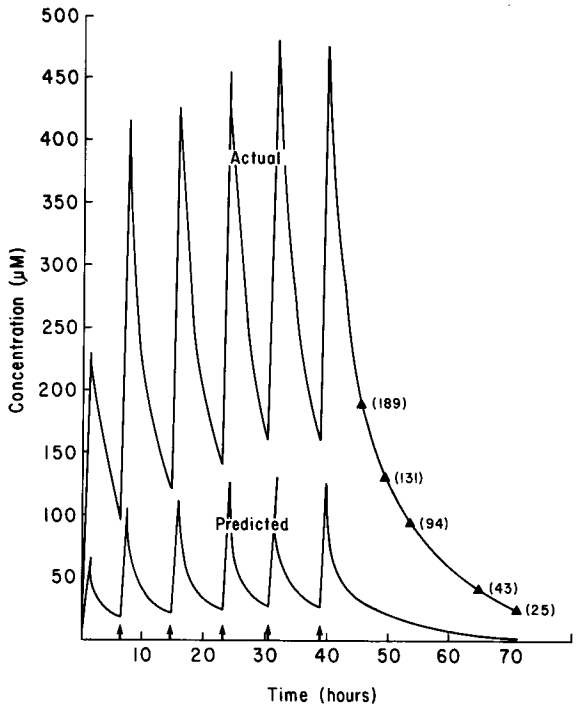


Figure 1. Acyclovir concentration profiles. The *upper profile* of acyclovir concentrations is reconstructed from actual values from the patient. The *lower profile* is based on average pharmacokinetic parameters adjusted for the patient's measured creatinine clearance.

Discussion

Neurotoxicity associated with acyclovir therapy in subjects infected with herpesviruses is an uncommon, but well described, adverse effect [5–10]. Manifestations include a combination of disorientation, hallucinations, delirium, tremors, ataxia, myoclonic jerks, slurred speech, seizures, or other motor or mental disturbances. Of the 22 reported cases, including ours, (a complete list of case reports is available upon request from S.F.) neurotoxicity has been reversible. Our case is unusual in that the patient's acyclovir clearance was much lower and serum concentration much higher than would have been predicted from the levels of BUN and serum creatinine [11]. Although the dosage of 500 mg/m² (~16.8 mg/kg) every 8 h was twice that usually used for herpes simplex virus infections [5, 6], it is used in the treatment of varicella-zoster virus infections [1].

Drug-related neurotoxicity has occurred in subjects receiving a wide range of doses and has occurred, in some instances, in subjects receiving doses lower than that received by our patient [5, 6]. Patients with normal renal function, who received the same dosage as our patient did, should have peak and trough concentrations of 90–100 μ*M* and <10 μ*M*, respectively [11]. The measured values were, however, substantially higher. Based on the pharmacokinetic analysis of the measured concentrations, the pattern of accumulation for the entire dose regimen has been reconstructed [4] (figure 1). The estimated peak concentrations were 290 μ*M* after the initial dose and 470 μ*M*

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after the sixth dose. The predose concentration (trough) before the last dose was 160 μ*M*, 15–16-fold higher than the expected value.

Elimination of acyclovir is primarily renal and involves both tubular secretion and glomerular filtration [11]. The actual acyclovir clearance, determined from the measured serum concentration, was 27 mL/min per 1.73 m² when the serum creatinine was 1.5 mg/dL. The acyclovir clearance is much lower than would be predicted, even when adjusted for impaired renal function, and is similar to the mean clearance of 29 ml/min per 1.73 m² reported for patients with end-stage renal disease [12]. Two days after stopping acyclovir therapy and when the serum creatinine was 1.2 mg/dL, the measured creatinine clearance was 27 mL/min per 1.73 m². The calculated creatinine clearance [13] based on this measured serum creatinine would have been 71 mL/min per 1.73 m². This latter value would not warrant dosage modification [11]. The results from measured creatinine clearance indicated, however, that the patient's dosage should have been reduced to twice daily.

To illustrate the disparity between measured drug concentrations and what would be predicted from average pharmacokinetic parameters, we produced a simulation [11, 12] with elimination adjusted for the apparent creatinine clearance of this patient (figure 1). The patient's measured acyclovir concentrations were five to 15 times higher than their predicted values. Impaired hepatic function, indicated by the elevated level of bilirubin, may have been a contributing factor to the very low systemic acyclovir clearance.

Of the 21 previously reported cases of acyclovir neurotoxicity, drug concentrations were measured in eight subjects [8–10]. In one series, elevated "peak" values (4 h postinfusion, 130–268 μ*M*) were found in four of six patients with neurotoxicity [8]. In these patients the temporal relation between the acyclovir concentration and the occurrence of CNS dysfunction was not discussed. In another patient it was noted that the extrapolated peak concentrations would have been "very high" [9]. Wade and Meyers [8] proposed a predisposition to neurotoxicity due to CNS insults from chemotherapy, irradiation, and herpesvirus infections as one possible explanation for the adverse effects of acyclovir. An alternative explanation is illustrated by our patient's unrecognized decreased clearance of acyclovir and concomitantly elevated concentrations of the drug. Her creatinine and acyclovir clearances were disproportionately decreased relative to serum creatinine. In retrospect, several factors suggested that there would be difficulty in eliminating the drug. The patient had structural abnormalities of the kidney and had received repeated courses of nephrotoxic drugs (e.g., cisplatin, cyclophosphamide, aminoglycosides, etc.). Of the 22 reported cases (including ours), 17 (77%) patients had an underlying malignancy or were organ transplant recipients. In most instances it was indicated that the patients had received nephrotoxic drugs. Presumably, detailed histories would have revealed that all these patients had received drugs known to impair renal function. Of the remaining five subjects, four had renal failure. Thus, renal dysfunction that causes impaired drug clearance and produces elevated acyclovir concentration increases the risk for neurotoxicity.

In summary, we present a patient with acyclovir neurotoxicity associated with extremely high serum concentrations, with a marked disparity between actual acyclovir and creatinine clearances and values predicted from the serum creatinine. We attributed this disparity, in part, to the toxic drugs the patient had received for her underlying malignancy, which produced sufficient renal dysfunction to cause poor clearance of acyclovir. The hepatic disease may have further contributed to decreased clearance. The role of the morphine sulfate that the patient received is unclear. A previous report [9] attributed the neurotoxicity to a meperidine-acyclovir interaction; however, the patient had impaired renal function. We recommend measuring serum concentrations of acyclovir in those patients whose history suggests exposure to nephrotoxic drugs or other factors that can impede renal function. Peak levels of acyclovir of 40–80 μ*M* and trough levels of <10 μ*M* (at a dosage of 750 mg/m² per d) are efficacious, well above the sensitivity of the virus, and below the apparent neurotoxic concentrations [5, 6]. Measuring acyclovir in serum is usually not possible. Presently, a measured, rather than a calculated, creatinine clearance is indicated as an alternative means for determining the dose of acyclovir to be used to treat patients at risk for decreased drug clearance. It is noteworthy that many of the patients receiving intravenous acyclovir are those at highest risk to have decreased clearance of the drug (e.g., organ transplant recipients, patients with cancer, neonates, and the elderly). Finally, additional studies are needed to more precisely define the relation between clinical efficacy, serum concentrations, and sensitivities of the isolated virus.

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